

# Mathematics for Computer Science – 2

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# Definition of congruence (同余) .

“同余” 这个概念是 CSers 的好朋友。

## Definition

We say that  $a$  is congruent to  $b$  modulo  $m$ , and we write

$$a \equiv b \pmod{m},$$

if  $m$  divides  $a - b$ . The number  $m$  is called the *modulus* (模数) of the congruence.

## Some notations.

$$a \equiv b \pmod{m} \iff$$

- $\exists k \in \mathbb{Z}, a = km + b.$
- $(a \bmod m) = (b \bmod m)$
- $m \mid (a - b)$

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提示.

表达是思维的指针。

## Example

$$26 \equiv 8 \pmod{9} \quad \text{and} \quad 6 \equiv 55 \pmod{7},$$

*since*

$$9 \mid (26 - 8) \quad \text{and} \quad 7 \mid (6 - 55),$$

*or, equivalently:*

$$8 = 26 - 2 \cdot 9 \quad \text{and} \quad 55 = 6 + 7 \cdot 7.$$

## Lemma

*If  $a_1 \equiv b_1 \pmod{m}$  and  $a_2 \equiv b_2 \pmod{m}$ , then*

$$a_1 \pm a_2 \equiv b_1 \pm b_2 \pmod{m}$$

*and*

$$a_1 a_2 \equiv b_1 b_2 \pmod{m}$$

课堂练习!

## 练习

假设  $x \equiv a \pmod{n}$ , 且  $n = p \cdot q$ ,  $p, q$  是两个素数。请证明, 因此必然有  $x \equiv a \pmod{p}$  和  $x \equiv a \pmod{q}$ 。



## Examples

Since  $10000 \equiv 1 \pmod{3}$  and  $998 \equiv 2 \pmod{3}$ , then

$$10000 \cdot 998 \equiv 2 \pmod{3}.$$

## 练习

计算以下等式：

$$7^{500} \bmod 8$$

$$1! + 2! + 3! + \cdots + 99! + 100! \bmod 12$$

$$2^{20} - 1 \bmod 41$$

## 练习

计算以下等式：

$$7^{500} \bmod 8$$

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$$2^{20} - 1 \bmod 41$$

## 答案

1、9、0。

# Properties of Congruence.

Negative number.

Let  $x$  and  $n$  be two positive integers and  $x < n$ , what does  $-x \bmod n$  mean?

# Properties of Congruence.

## Negative number.

Let  $x$  and  $n$  be two positive integers and  $x < n$ , what does  $-x \bmod n$  mean?

## Intuition.

Consider that the negative of  $x$  is the number  $x'$  such that  $x + x' = 0$ , that is:

$$x + x' \equiv 0 \pmod{n}.$$

Since  $x < n$ , hence  $x' = n - x$ .

# Two's Complement(二进制补码)

## Two's Complement

A signed number represented in  $n$  bits. The range of the numbers is  $[-2^{n-1}, 2^{n-1} - 1]$ , and the rule is described as follows:

- Positive integers, in the range 0 to  $2^{n-1} - 1$ , are stored in regular binary form. The sign bit is set to 0.
- Negative integers  $-x$ , with  $1 \leq x \leq 2^{n-1}$ , are calculated by first constructing  $x$  in binary, then inverting all the bits of  $x$  and finally adding 1. The sign bit is set to 1.

# Two's Complement

Another way to remember two's complement.

The negative of  $x$  equals  $2^n - x$ . Since  $2^n - x = \underbrace{111 \cdots 11}_n + 1 - x$ ,  
and  $\underbrace{111 \cdots 11}_n - x$  is the same as inverting all the bits of  $x$ .

## Theorem (Cancellation Law.)

*If  $\gcd(c, m) = 1$  and*

$$ac \equiv bc \pmod{m},$$

*then*

$$a \equiv b \pmod{m},$$

*where gcd shorts for the greatest common divisor.*



# Cancellation Law.

## Proof.

By definition of congruence, we have  $m|(ac - bc)$ , equivalently,  $m|(a - b)c$ . Since  $\gcd(c, m) = 1$ , it follows that  $m \mid (a - b)$ , so as claimed.



Another perspective.

If  $\gcd(c, m) = 1$  then  $\exists r, s \in \mathbb{Z}$  s.t.

$$rc + sm = 1$$

both sides of the equation modulo  $m$ , we have:

$$rc \equiv 1 \pmod{m}$$

means  $r$  is the multiplicative inverse(乘法逆元) of  $c \bmod m$ , let it be  $c^{-1}$ .

## Partially solve the congruence $ax \equiv b \pmod{m}$ .

### Example

*To solve  $3x \equiv 2 \pmod{11}$ .*

*Firstly, by using egcd algorithm, to compute that  $3^{-1} = 4$ , because  $3 \cdot 4 \equiv 1 \pmod{11}$ . Multiply 4 to the equation and obtain*

$$x \equiv 8 \pmod{11}.$$

## 问题.

消去律源自于同余的定义与欧几里得引理。可以执行消去操作是因为存在唯一乘法逆元，乘法逆元源自于 Bezout 定理，但是 Bezout 系数不唯一。请问，为什么乘法逆元会唯一呢？

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## 回答.

需要重新审视 Bezout 系数的形式。

# Properties of Congruence.

## Lemma

*For  $n \in \mathbb{N}$ , congruence modulo  $n$  forms an equivalence relation(等价关系) of  $\mathbb{Z}$ .*

## Proof.

It is easy to check that:

1. Reflexive(自反性).  $a \equiv a \pmod{n}$
2. Symmetric(对称性). If  $a \equiv b \pmod{n}$  then  $b \equiv a \pmod{n}$
3. Transitive(传递性). If  $a \equiv b \pmod{n}$  and  $b \equiv c \pmod{n}$  then  $a \equiv c \pmod{n}$  □

## 问题

等价关系意味着什么？

## 问题

等价关系意味着什么？

## 条件反射式回答.

存在等价类，即对数据的划分 (Partition)。



# Equivalence relation and equivalence classes.

## Definition

*When a set  $\mathbb{S}$  has an equivalence relation on it, then the equivalence relation partitions (划分、分割) the set  $\mathbb{S}$  into disjoint subsets, called equivalence classes (等价类), defined by the property that two elements are in the same equivalence class if they are equivalent.*

# Congruence classes modulo $m$ .

The set of congruence classes modulo  $m$  is denoted by  $\mathbb{Z}/m\mathbb{Z}$ .

There are exactly  $m$  congruence classes in  $\mathbb{Z}/m\mathbb{Z}$ . That is :

$$\mathbb{Z}/m\mathbb{Z} = \{[0]_m, [1]_m, \dots, [m-1]_m\}.$$

## Example

*When  $m = 2$ ,  $\mathbb{Z}/2\mathbb{Z} = \{[0]_2, [1]_2\}$ . The congruence class  $[1]_2$  is the set of all integers congruent to 1 modulo 2. Thus  $[1]_2$  is the set of all odd integers. Similarly, the congruence class  $[0]_2$  is the set of all even integers.*

# Proposition of Congruence.

## Proposition

*If  $[a_1]_m = [a_2]_m$  and  $[b_1]_m = [b_2]_m$ , then*

$$[a_1 \pm b_1]_m = [a_2 \pm b_2]_m, \quad \text{and} \quad [a_1 b_1]_m = [a_2 b_2]_m.$$

## Proof.

It is easy. Transform the form  $[a]_m = [b]_m$  to  $a \equiv b \pmod{m}$  and use Proposition 2.2. □

# Notations of Congruence classes modulo $m$ .

- Any element  $b$  of a congruence class  $[a]_m$  is called a *representative* (代表元) of that class.
- The set of all the least nonnegative representative of  $\mathbb{Z}/m\mathbb{Z}$  is the set of integers  $\{0, 1, 2, \dots, m-1\}$ , that is called the *least residue system* modulo  $m$  (模  $m$  的最小剩余系).
- Any set of  $m$  integers, no two of which are congruent modulo  $m$ , is called a *complete residue system* modulo  $m$  (模  $m$  的完全剩余系).

# Notations of Congruence classes modulo $m$ .

## Example

*Let  $m = 7$ , the least residue system modulo  $m$  is the set  $\{0, 1, 2, 3, 4, 5, 6\}$ , and a complete residue system modulo  $m$  may be the set  $\{14, 8, 23, 46, 61, 13\}$ .*

In this section, we focus on modular exponentiation which is an important arithmetic primitive. Its task is that given integers  $x$ ,  $y$  and  $m$  to compute

$$x^y \bmod m.$$

## Example

*To compute  $2^{16} \bmod 11$ . We compute:*

$$2^2 \bmod 11 = 4$$

$$2^4 \bmod 11 = 4 \cdot 4 \bmod 11 = 5$$

$$2^8 \bmod 11 = 5 \cdot 5 \bmod 11 = 3$$

$$2^{16} \bmod 11 = 3 \cdot 3 \bmod 11 = 9$$

## Example

*To compute  $2^{22} \bmod 11$ . We compute:*

$$2^2 \bmod 11 = 4$$

$$2^4 \bmod 11 = 4 \cdot 4 \bmod 11 = 5$$

$$2^8 \bmod 11 = 5 \cdot 5 \bmod 11 = 3$$

$$2^{16} \bmod 11 = 3 \cdot 3 \bmod 11 = 9$$

*then*

$$(2^{22} = 2^2 \cdot 2^4 \cdot 2^{16}) \equiv 4 \cdot 5 \cdot 9 \equiv 4 \pmod{11}$$



The process can be expressed as a recursive form, by that, we sharply improve the efficiency from performing  $O(y)$  multiplications to  $O(\log(y))$ .

$$x^y = \begin{cases} (x^{\lfloor y/2 \rfloor})^2 & \text{if } y \text{ is even;} \\ x \cdot (x^{\lfloor y/2 \rfloor})^2 & \text{if } y \text{ is odd.} \end{cases} \quad (1)$$

## Listing 1: Recursive Modular Exponentiation

```
1  # Recursive Function to calculate  
2  #  $(x^y) \% p$  in  $O(\log y)$   
3  def rec_mod_exp(x, y, p):  
4      if (y == 0): return 1  
5      z = rec_mod_exp(x, y//2, p)  
6      if ((y & 1) == 0): #y is an even number  
7          return z*z % p  
8      else: #y is an odd number  
9          return x*z*z % p
```

# Mod Exponentiation: from recursive to iterative.

We describe how to transform the recursive algorithm to an iterative algorithm as follows. Firstly, we treat integer  $y$  as a polynomial (or a binary string):

$$y = y_{n-1}2^{n-1} + y_{n-2}2^{n-2} \cdots + y_12 + y_0,$$

where  $y_i \in \{0, 1\}$ .

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Secondly, transform  $x^y$  as:

$$x^y = \prod_{i=0}^{n-1} x^{y_i 2^i}$$

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where  $y_i \in \{0, 1\}$ .

Secondly, transform  $x^y$  as:

$$x^y = \prod_{i=0}^{n-1} x^{y_i 2^i}$$

Finally, start with  $x$  and repeatedly square modulo  $m$ , multiply the terms with  $y_i = 1$  and get the result.

## Example

Let  $x = 7$ ,  $y = 10$ ,  $m = 11$ , to compute  $x^y \bmod m$ . The binary string of  $y$  is 1010, thus we compute:

$$y_0 = 0, \quad x^{2^0} \equiv 7 \pmod{m}$$

$$y_1 = 1, \quad x^{2^1} \equiv 5 \pmod{m}$$

$$y_2 = 0, \quad x^{2^2} \equiv 3 \pmod{m}$$

$$y_3 = 1, \quad x^{2^3} \equiv 9 \pmod{m}$$

Then, multiply the terms with  $y_i = 1$ , we have  
 $x^y = (5 \cdot 9) \bmod 11 = 1$

练习.

Let  $x = 5$ ,  $y = 13$ ,  $m = 13$ , to compute  $x^y \bmod m$ .

练习.

Let  $x = 5$ ,  $y = 13$ ,  $m = 13$ , to compute  $x^y \bmod m$ .

答案.

5.



# Mod Exponentiation (Iterative version).

Because the white board is too narrow to show the code, so it is your home work.

# Some topics using modular arithmetic.

Our following job is to play with number using modular arithmetic, and find some patterns or rules.

# Find the patterns.

Let  $p = 7$ , and for every  $1 \leq a < p$ , compute  $a^i \bmod p$ , where  $1 \leq i < p$ . We have:

| $a$ | $a^2$ | $a^3$ | $a^4$ | $a^5$ | $a^6$ |
|-----|-------|-------|-------|-------|-------|
| 1   | 1     | 1     | 1     | 1     | 1     |
| 2   | 4     | 1     | 2     | 4     | 1     |
| 3   | 2     | 6     | 4     | 5     | 1     |
| 4   | 2     | 1     | 4     | 2     | 1     |
| 5   | 4     | 6     | 2     | 3     | 1     |
| 6   | 1     | 6     | 1     | 6     | 1     |

# Find the patterns.

Let  $p = 7$ , and for very  $1 \leq a < p$ , compute  $a^i \bmod p$ , where  $1 \leq i < p$ . We have:

| $a$ | $a^2$ | $a^3$ | $a^4$ | $a^5$ | $a^6$ |
|-----|-------|-------|-------|-------|-------|
| 1   | 1     | 1     | 1     | 1     | 1     |
| 2   | 4     | 1     | 2     | 4     | 1     |
| 3   | 2     | 6     | 4     | 5     | 1     |
| 4   | 2     | 1     | 4     | 2     | 1     |
| 5   | 4     | 6     | 2     | 3     | 1     |
| 6   | 1     | 6     | 1     | 6     | 1     |

May you find some patterns?

## More data to find the patterns.

Let  $p = 11$ , and for every  $1 \leq a < p$ , compute  $a^i \bmod p$ , where  $1 \leq i < p$ . We have:

| $a$ | $a^2$ | $a^3$ | $a^4$ | $a^5$ | $a^6$ | $a^7$ | $a^8$ | $a^9$ | $a^{10}$ |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| 1   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1        |
| 2   | 4     | 8     | 5     | 10    | 9     | 7     | 3     | 6     | 1        |
| 3   | 9     | 5     | 4     | 1     | 3     | 9     | 5     | 4     | 1        |
| 4   | 5     | 9     | 3     | 1     | 4     | 5     | 9     | 3     | 1        |
| 5   | 3     | 4     | 9     | 1     | 5     | 3     | 4     | 9     | 1        |
| 6   | 3     | 7     | 9     | 10    | 5     | 8     | 4     | 2     | 1        |
| 7   | 5     | 2     | 3     | 10    | 4     | 6     | 9     | 8     | 1        |
| 8   | 9     | 6     | 4     | 10    | 3     | 2     | 5     | 7     | 1        |
| 9   | 4     | 3     | 5     | 1     | 9     | 4     | 3     | 5     | 1        |
| 10  | 1     | 10    | 1     | 10    | 1     | 10    | 1     | 10    | 1        |

## 猜想

For  $p$  is a prime, and  $\gcd(a, p) = 1$ ,

$$a^{p-1} \equiv 1 \pmod{p}$$

## Another computation.

$\forall 1 < a < p$ , compute  $a \cdot i \bmod p$ , for  $1 \leq i < p$ . For example, let  $a = 2$ ,  $p = 7$ , we have:

| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|-------------|---|---|---|---|---|---|
| $a = 1$     | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 2$     | 2 | 4 | 6 | 1 | 3 | 5 |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |

## Another computation.

Continue the computation...

|             |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|
| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 1$     | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 2$     | 2 | 4 | 6 | 1 | 3 | 5 |
| $a = 3$     | 3 | 6 | 2 | 5 | 1 | 4 |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |



# Another computation.

Continue the computation...

|             |   |   |   |   |   |   |
|-------------|---|---|---|---|---|---|
| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 1$     | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 2$     | 2 | 4 | 6 | 1 | 3 | 5 |
| $a = 3$     | 3 | 6 | 2 | 5 | 1 | 4 |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |
|             |   |   |   |   |   |   |

Have a decision?

# Another computation.

Finally!

| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|-------------|---|---|---|---|---|---|
| $a = 1$     | 1 | 2 | 3 | 4 | 5 | 6 |
| $a = 2$     | 2 | 4 | 6 | 1 | 3 | 5 |
| $a = 3$     | 3 | 6 | 2 | 5 | 1 | 4 |
| $a = 4$     | 4 | 1 | 5 | 2 | 6 | 3 |
| $a = 5$     | 5 | 3 | 1 | 6 | 4 | 2 |
| $a = 6$     | 6 | 5 | 4 | 3 | 2 | 1 |

# Conjecture.

The trick is,  $p$  is a prime number! We conjecture: if  $p$  is a prime,  $\forall a$  which is not divided by  $p$ ,  $a \cdot i \bmod p$ , for  $1 \leq i < p$ , is a permutation of numbers from 1 to  $p - 1$ .

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$$a, 2a, 3a, \dots, (p-1)a \pmod{p}$$

are the same as the numbers:

$$1, 2, 3, \dots, p-1$$

although they may be in a different order.

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$$a, 2a, 3a, \dots, (p-1)a \pmod{p}$$

are the same as the numbers:

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although they may be in a different order.

$$S = \{a \cdot i \bmod p, 1 \leq i < p\}$$

is also called a complete system of residues modulo  $p$ .

# Proof by contradiction.

Of course, we need a proof!

# Proof by contradiction.

Of course, we need a proof!

Proof.

Proof by contradiction (informal and incomplete). If we are wrong, then there exist  $i$  and  $j$  such that,

$$a \cdot i \equiv a \cdot j \pmod{p}$$

where  $i \neq j$ . However, then we can cancel the  $a$  from the equation! (Cancellation Law.) □

# Do a simple job!

Multiply all  $1 \leq i < p$ , and all  $a \cdot i \bmod p$ , we have:

$$\prod_{i=1}^{p-1} i = \prod_{i=1}^{p-1} (a \cdot i \bmod p)$$



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Convince yourself:

$$\prod_{i=1}^{p-1} i \equiv \prod_{i=1}^{p-1} a \cdot i \equiv a^{p-1} \prod_{i=1}^{p-1} i \pmod{p}$$

# Do a simple job!

Multiply all  $1 \leq i < p$ , and all  $a \cdot i \bmod p$ , we have:

$$\prod_{i=1}^{p-1} i = \prod_{i=1}^{p-1} (a \cdot i \bmod p)$$

Convince yourself:

$$\prod_{i=1}^{p-1} i \equiv \prod_{i=1}^{p-1} a \cdot i \equiv a^{p-1} \prod_{i=1}^{p-1} i \pmod{p}$$

Cancel the big number, we have:

$$a^{p-1} \equiv 1 \pmod{p}$$

# Fermat's little theorem.

## Theorem (Fermat's little theorem.)

*Let  $p$  be a prime number, and let  $a$  be any number with  $a \not\equiv 0 \pmod{p}$ . Then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

# An Exercise using Fermat's little theorem.

## Exercise.

Let  $p = 17$  be a prime number, and let  $a = 2$ , what is  $a^{2024} \bmod p$ ?

# An Exercise using Fermat's little theorem.

## Exercise.

Let  $p = 17$  be a prime number, and let  $a = 2$ , what is  $a^{2024} \bmod p$ ?

$$2^{2024} \equiv 2^{126 \cdot 16 + 8} \equiv 2^8 \equiv 1 \pmod{17}.$$

# Exercises using Fermat's little theorem.

## Exercises.

- $a \equiv 9^{794} \pmod{73}$ , find  $a$  and  $0 \leq a \leq 73$ .
- Solve  $x^{87} \equiv 6 \pmod{29}$ .

# More computation.

If the modulus is a composite number, then our trick will fail! For example, let  $n = 6$ ,

| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 |
|-------------|---|---|---|---|---|
| $a = 1$     | 1 | 2 | 3 | 4 | 5 |
| $a = 2$     | 2 | 4 | 0 | 2 | 4 |
| $a = 3$     | 3 | 0 | 3 | 0 | 3 |
| $a = 4$     | 4 | 2 | 0 | 4 | 2 |
| $a = 5$     | 5 | 4 | 3 | 2 | 1 |

# One more computation.

Let  $n = 9$ ,

| $a \cdot i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|-------------|---|---|---|---|---|---|---|---|
| $a = 1$     | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| $a = 2$     | 2 | 4 | 6 | 8 | 1 | 3 | 5 | 7 |
| $a = 3$     | 3 | 6 | 0 | 3 | 6 | 0 | 3 | 6 |
| $a = 4$     | 4 | 8 | 3 | 7 | 2 | 6 | 1 | 5 |
| $a = 5$     | 5 | 1 | 6 | 2 | 7 | 3 | 8 | 4 |
| $a = 6$     | 6 | 3 | 0 | 6 | 3 | 0 | 6 | 3 |
| $a = 7$     | 7 | 5 | 3 | 1 | 8 | 6 | 4 | 2 |
| $a = 8$     | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |



## 观察.

If  $n$  is a composite number, the numbers

$$a, 2a, 3a, \dots, (n-1)a \pmod{n}$$

may **NOT** be the same as the numbers:

$$1, 2, 3, \dots, n-1$$

## Check the observation.

Let  $n = 9$ , for all  $i \in [1, n - 1]$ , and  $\gcd(i, n) = 1$ :

| $a \cdot i$ | 1 | 2 | 4 | 5 | 7 | 8 |
|-------------|---|---|---|---|---|---|
| $a = 1$     | 1 | 2 | 4 | 5 | 7 | 8 |
| $a = 2$     | 2 | 4 | 8 | 1 | 5 | 7 |
| $a = 4$     | 4 | 8 | 7 | 2 | 1 | 5 |
| $a = 5$     | 5 | 1 | 2 | 7 | 8 | 4 |
| $a = 7$     | 7 | 5 | 1 | 8 | 4 | 2 |
| $a = 8$     | 8 | 7 | 5 | 4 | 2 | 1 |

# Conjecture.

We conjecture: Let  $n$  be a composite number, denotes

$$S = \{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}$$

Then  $\forall a$  with  $\gcd(a, n) = 1$ , denotes

$$S' = a \cdot S \pmod{n}$$

we have:

$$S = S'$$

## Euler's phi function.

Define:

$$\varphi(n) = |\{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}|$$

The function  $\varphi$  is called *Euler's phi function*.

Then:

$$S = \{b_1, b_2, \dots, b_{\varphi(n)} : 1 \leq b_i < n \text{ and } \gcd(b_i, n) = 1\}$$

$$S' = \{a \cdot b_1, a \cdot b_2, \dots, a \cdot b_{\varphi(n)} \pmod{n} : \\ b_i \in S \text{ and } \gcd(a, n) = 1\}.$$

To prove

$$S = S'$$

Check that if there exist:

$$a \cdot b_i \equiv a \cdot b_j \pmod{n}$$

where  $b_i \neq b_j$ . Then by Cancellation Law,

$$b_i \equiv b_j \pmod{n}.$$

Contradiction!

# Do a similar simple job!

Multiply all the numbers in  $S$  and  $S'$ , we have:

$$\prod_{i=1}^{\varphi(n)} b_i = \prod_{i=1}^{\varphi(n)} (a \cdot b_i \bmod n)$$

$$\prod_{i=1}^{\varphi(n)} b_i \equiv \prod_{i=1}^{\varphi(n)} a \cdot b_i \pmod{n}$$

$$\prod_{i=1}^{\varphi(n)} b_i \equiv a^{\varphi(n)} \prod_{i=1}^{\varphi(n)} b_i \pmod{n}$$

Cancel the big number, we have:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

# Euler's Theorem.

(Euler's Theorem.)

Let  $n$  be a positive composite number,  $a$  be a positive integer with  $\gcd(a, n) = 1$ , then:

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$



例.

设  $n = 100$ , 则  $\varphi(n) = 40$ , 所以  $7^{40} \equiv 1 \pmod{n}$ 。同样,  $9^{40} \equiv 1 \pmod{n}$ 。

# Example.

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问题.

为什么  $\varphi(n) = 40$  ? 或者, 给定  $n$ , 如何求  $\varphi(n)$  ?

# More about Euler's Phi Function.

## Definition (Euler's Phi Function.)

*Define:*

$$\varphi(n) = |\{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}|$$

*The function  $\varphi$  is called Euler's phi function.*

## Observations

$\varphi(p) = p - 1$ , where  $p$  is a prime.

$\varphi(p^k) = p^k - p^{k-1}$ , where  $p$  is a prime.

# More about Euler's Phi Function..

Question.

How to compute  $\varphi(m)$  where  $m = p^i q^j$  with  $p$  and  $q$  are prime.

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A related question.

How to compute  $\varphi(mn)$  where  $\gcd(m, n) = 1$ .

# More about Euler's Phi Function.

Compute  $\varphi(mn)$

Display the positive integers not exceeding  $mn$  in the following way.

|         |         |          |         |                |
|---------|---------|----------|---------|----------------|
| 1       | $m + 1$ | $2m + 1$ | $\dots$ | $(n - 1)m + 1$ |
| 2       | $m + 2$ | $2m + 2$ | $\dots$ | $(n - 1)m + 2$ |
| 3       | $m + 3$ | $2m + 3$ | $\dots$ | $(n - 1)m + 3$ |
| $\dots$ | $\dots$ | $\dots$  | $\dots$ | $\dots$        |
| $r$     | $m + r$ | $2m + r$ | $\dots$ | $(n - 1)m + r$ |
| $\dots$ | $\dots$ | $\dots$  | $\dots$ | $\dots$        |
| $m$     | $2m$    | $3m$     | $\dots$ | $mn$           |

# More about Euler's Phi Function.

## Basic idea.

Find all the elements which are relatively prime to both  $n$  and  $m$ , then it is relatively prime to  $mn$ . Formally:

$$\forall a, \gcd(a, n) = 1 \text{ and } \gcd(a, m) = 1 \implies \gcd(a, mn) = 1.$$

# More about Euler's Phi Function.

## Counting

- How many rows satisfy  $\gcd(r, m) = 1$ ? Ans :  $\varphi(m)$ . Note that, if  $\gcd(r, m) = 1$  then  $\gcd(km + r, m) = 1$ , for  $k \in [0..n - 1]$ .
- At  $r$ th row, how many integers have  $\gcd(km + r, n) = 1$ , for  $k \in [0..n - 1]$ ? Ans:  $\varphi(n)$ . Note that, we have  $\gcd(n, m) = 1$ .
- Hence, there are  $\varphi(m)$  rows, each containing  $\varphi(n)$  integers relatively prime to  $mn$ .



# More about Euler's Phi Function.

Remark (Why the conclusion holds?)

*If  $\gcd(a, m) = 1$  and  $\gcd(a, n) = 1$  then  $\gcd(a, mn) = 1$ .*

# More about Euler's Phi Function.

## Remark (Why the second item holds?)

*The elements in  $r$ th row are:  $r, m + r, \dots, (n - 1)m + r$  with  $\gcd(m, n) = 1$ .  $\forall k_i \neq k_j, k_i m + r \not\equiv k_j m + r \pmod{n}$ . Otherwise,  $k_i = k_j$  by our Golden Law (Cancellation Law), contradiction! It means the  $n$  elements in  $r$ th row form "a complete system of residues modulo  $n$ ", that is  $\{r, m + r, \dots, (n - 1)m + r\} \pmod{n} = \{0, 1, 2, \dots, n - 1\}$ . Hence, exactly  $\varphi(n)$  of these integers are relatively prime to  $n$ .*

## Theorem

*Let  $m$  and  $n$  be relatively prime positive integers. Then  $\varphi(mn) = \varphi(m)\varphi(n)$ .*

# More about Euler's Phi Function.

Some easy generalizations.

How to relate  $\varphi(mn) = \varphi(m)\varphi(n)$  where  $\gcd(m, n) = 1$  with  $\varphi(m)$  where  $m = p^i q^j$  ?

# More about Euler's Phi Function.

## Some easy generalizations.

How to relate  $\varphi(mn) = \varphi(m)\varphi(n)$  where  $\gcd(m, n) = 1$  with  $\varphi(m)$  where  $m = p^i q^j$  ?

Ans:  $\gcd(p^i, q^j) = 1$  when  $p$  and  $q$  are relatively prime. Hence  $\varphi(p^i q^j) = \varphi(p^i)\varphi(q^j)$ .

# More about Euler's Phi Function.

## Some easy generalizations.

How to relate  $\varphi(mn) = \varphi(m)\varphi(n)$  where  $\gcd(m, n) = 1$  with  $\varphi(m)$  where  $m = p^i q^j$  ?

Ans:  $\gcd(p^i, q^j) = 1$  when  $p$  and  $q$  are relatively prime. Hence  $\varphi(p^i q^j) = \varphi(p^i)\varphi(q^j)$ .

How to generalize the result to  $\varphi(m)$  where  $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$  ?

# More about Euler's Phi Function.

## Some easy generalizations.

How to relate  $\varphi(mn) = \varphi(m)\varphi(n)$  where  $\gcd(m, n) = 1$  with  $\varphi(m)$  where  $m = p^i q^j$  ?

Ans:  $\gcd(p^i, q^j) = 1$  when  $p$  and  $q$  are relatively prime. Hence  $\varphi(p^i q^j) = \varphi(p^i)\varphi(q^j)$ .

How to generalize the result to  $\varphi(m)$  where  $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$  ?

Ans: Induction! Left as an exercise.

## 习题.

求  $a$ , 使得  $a$  满足以下条件:

- $a \equiv 7^{2003} \pmod{3750}$
- $1 \leq a \leq 5000$
- $a$  不被 7 整除.



## 习题.

使用费尔马小定理求解同余方程  $x^{51} \equiv 2 \pmod{17}$  和  $x^{50} \equiv 2 \pmod{17}$ 。

## 习题.

请求解  $x^{113} \equiv 2 \pmod{221}$ 。[提示: 113 与  $\varphi(221)$  互素。]

Prove the following theorem.

## Theorem (Euler's Phi Function.)

*Let  $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ , for all  $p_i$  ( $1 \leq i \leq k$ ) is a prime, then  $\varphi(m) = m(1 - 1/p_1)(1 - 1/p_2) \cdots (1 - 1/p_k)$ .*

- $pk, sk \leftarrow KG(\lambda)$ ,  $sk$  is local and secret, while  $pk$  is public.
- Given plaintext  $m$ , compute ciphertext  $c \leftarrow Enc(m, pk)$
- Given ciphertext  $c$ , compute plaintext  $m \leftarrow Dec(c, sk)$

- $pk, sk \leftarrow KG(\lambda)$ :
  - Choose two large primes  $p$  and  $q$ ,  $n = p \cdot q$ , let  $\varphi(n) = (p-1)(q-1)$ .
  - Choose a integer  $e$ , with  $\gcd(e, \varphi(n)) = 1$ , compute  $d$ , s.t.  $ed \equiv 1 \pmod{\varphi(n)}$ .
  - Let  $pk$  be  $e$  and  $n$ ;  $sk$  be  $d$  and  $n$ .
- $Enc(m, pk)$ : let  $c \equiv m^e \pmod{n}$
- $Dec(c, sk)$ : let  $m \equiv c^d \pmod{n}$

# Why RSA is correct?

Because of Euler's Theorem.

- $KG(\lambda)$ :
  - $e$  and  $d$  satisfy  $ed \equiv 1 \pmod{\varphi(n)}$ .
  - means:  $\exists k$ , s.t.  $ed = k\varphi(n) + 1$
- $Enc(m, pk)$ :  $c \equiv m^e \pmod{n}$
- $Dec(c, sk)$ :  $c^d \equiv m^{ed} \equiv m^{k\varphi(n)+1} \equiv m \pmod{n}$

- $KG(\lambda)$ :
  - Choose two primes  $p = 7$  and  $q = 11$ ,  $n = p \cdot q = 77$ , let  $\varphi(n) = (p - 1)(q - 1) = 60$ .
  - Choose a integer  $e = 7$ , with  $\gcd(e, \varphi(n)) = 1$ , compute  $d$ , s.t.  $ed \equiv 1 \pmod{\varphi(n)}$ ,  $d = 43$ .
  - Let  $pk$  be  $e = 7$  and  $n = 77$ ;  $sk$  be  $d = 43$  and  $n = 77$ .
- $Enc(m, pk)$ : Given  $m = 3$ , then  $c = 3^7 \bmod 77 = 31$ .
- $Dec(c, sk)$ : Let  $m = 31^{43} \bmod 77 = 3$ .

# What is a Concrete Introduction?

- Play with numbers.
- Find the patterns, find the fun.
- Programming is a good way to play.

# What have been covered?

- Congruence.
- Fermat's little theorem.
- Euler's theorem.



# What have been omitted?

- Fast multiplication.
- Powers: how to do fast power.

# What is the next step?

- From a new perspective to view Fermat's and Euler's theorems.
- From arithmetic go to algebra.